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IT mDV 26 002
Degree project 30 credits
February 2026

A Solar Internet

An intervention study

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Master's Programme in Computer Science



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Abstract

An increase in extreme weather and increased dependence on solar and wind power could lead to an energy infrastructure with more variability in capacity, one that periodically can't meet demand. That makes it important to understand intermittent infrastructures and how it could affect people's daily lives. In this study, I performed an intervention that simulated a scenario where the internet is run on solar power, such that the amount of bandwidth available was correlated to how strongly the sun shines. Through semi-structured interviews, this gave specific accounts of how people tried to adapt to intermittency. One significant obstacle was social connections and dependencies that made it hard for people to participate in the study. Insights into how to run a similar study and possible improvements are proposed.

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1. Introduction

Today, it is common to assume the internet should always be available everywhere at maximum capacity [23]. In a recent estimate, Information and Communications Technologies (ICT) make up between 2-4% of global carbon emissions, with few signs of being reduced [10]. Of the emissions attributed to ICTs, network infrastructure makes up a significant portion and have been estimated to emit more than data centres or computers [21]. A more recent report by the IEA predicts that emissions from data centres can double by 2024, with "AI" and cryptocurrencies as major drivers [14, p.8]. One response from HCI researchers is to focus on how the capacity of renewable energy sources like wind or solar power vary with the weather. The seasonal nature of these energy sources can mean that there isn't enough power to meet demand at all times of day. One can either try to eliminate this intermittency — with different battery storage and distribution mechanisms, or accept and adapt to it. My approach in this thesis is the latter.

There are reasons to consider what happens when the internet becomes less reliably available, when it becomes more intermittent. The first reason can be found in the approach used when designing the website for Low Tech Magazine to run on solar power [1]. In their efforts to "radically reduce the energy use associated with accessing [the magazine's] content" [1], they identified a tension between the intermittency in renewable power generation and the availability of the service. Increasing solar or battery capacity to improve availability gives diminishing returns, while significantly increasing the embodied energy requirements. Intermittency is therefore a relevant path to explore for lowering energy consumption.

Another reason is that as climate change progresses, extreme weather phenomena will become more common, which can affect infrastructure. Intermittency due to inadequate infrastructure is perhaps assumed to only happen in under-resourced countries, but that is not the case. For example, in Louisiana, USA, storm season has gotten more intense due to climate change, and old infrastructure designed for milder storms is regularly failing. Inhabitants have begun to live with the expectation that the infrastructure is intermittent [19]. As the weather will continue to become more extreme it therefore seems prudent to increase our understanding of intermittency.

Thirdly, adapting for high degrees of intermittency could increase resilience towards network outages. Using voluntary off-gridders as an example, Kris De Decker argues that an intermittent energy supply can lead to higher resilience towards power outages, so that each outage incurs minimal costs [7]. He contrasts this with an outage in a society with few outages, where machines and human skills are not adapted to intermittency. Such an outage would have enormous costs. The resilience of voluntary off-gridders is achieved through a combination of skills (e.g. hand washing clothes), adapted technology (e.g. better insulation in fridges), and adapted routines (e.g. cooking when solar power is available). If this argument is applicable for networking as well, it would mean that

adapting networking systems for increased degree of intermittency would yield benefits for societies today, in addition to societies in hypothetical futures. For example, if I was working on a shared document (in e.g. Google Docs) in a hypothetical intermittent future, it would be useful to be able to continue writing in it when my computer inevitably disconnects from the internet, and then sync it back with my collaborators when the computer reconnects. This scenario can easily be translated back to present day societies; my network provider might have trouble, or I simply wanted to write outside in a park. In this way, software that is adapted towards intermittency (e.g. [17]) could increase people's resilience towards disconnection.

However, despite these reasons, the underlying assumption of the ICT industry has been perpetual growth. Historically, even though the computer chips became denser and faster, the total resource use grew, following the Rebound Effect [13], and ICT's energy footprint continue to grow to this day [10]. Preist et al. calls the strategies and design ideals that builds on the assumption of continued exponential growth in the capacities of ICTs the Cornucopian Paradigm [23]. This paradigm continues to drive increased demand, embedding more and more services into everyday practice, as services that are at first deemed unnecessary eventually become essential.

Challenging this paradigm is what motivates this thesis, which investigates the following question: How can increasing the intermittency of the internet affect the everyday life of people?

Instead of studying how downstream changes (consumption patterns) affect the upstream (energy production), I aimed to study the effects on the downstream (consumption patterns) from changes in the upstream (internet availability). This is a way of challenging the Cornucopian paradigm from the side of production by limiting the output to not meet the demand all the time.

Following the tradition of Critical Design [25], I want to increase understanding of people's relationship to the constant availability of internet services. The goal of Critical Design is to critically examine underlying — often invisible — assumptions by surfacing those assumptions through a provocation, making them visible and open for examination and reflection. In this study, I have attempted this by designing an intervention that ties participants' access to the internet to how strong the sun shines at the moment. This removes participants' constant access to the internet, making their connection intermittent. Like the fan that you only notice when it turns off, my hope was that this would make some assumptions the participants held about the internet more visible to them.

1.1. Delimitations

Due to the time constraints of a master's thesis, I have chosen to work with a small number of participants, and the length of each intervention was constrained to two weeks. Similarly, the demographic of the participants was chosen for convenience, prioritising

groups that I had access to from my studies and prior voluntary work experience.

The main intervention into the participants' lives in this study is their modified access to the internet. This study focuses only on their connection and relationship to the internet at home, as that requires a relatively limited intrusion. Modifying the participants' mobile data or Wi-Fi connection outside the home would require installing software on all their devices. For home Wi-Fi, it is enough to put a pre-programmed router between the participants' router and the WAN-port.

2. Related Work

2.1. Intermittent supply

Previous research focus directly on intermittency, or areas related to it, such as disconnection from the internet. Topics range from the effects of intermittent power supply on infrastructure [24, 22], all the way to end-users' experience of intermittency [16]. Raghavan et al. consider a scenario where, in contrast to today, power is exclusively generated from sources like wind and solar power that depend on the weather conditions of the local geographical area. They assume that "energy supply will be significantly less than its potential demand", meaning that the availability at the point of consumption will be volatile. From there they outline how this would affect the internet infrastructure and various ways to mitigate the problems. One workaround is to use delay-tolerant networking [24]. For example, by sending data over multiple pathways in the network the likelihood that any one of them would be down due to lack of power is decreased [20], at the cost of increased energy consumption overall. By caching important data on the client machine, services that normally require an internet connection at the time-of-use (e.g. forums) can be made to work even when the machine is periodically offline [26]. Research on applications of a similar nature has recently been taken up under the banner of Local-first software [17]. The ambition being to create network-connected real-time collaborative software like Google Docs that still work if you lose your internet connection.

Some researchers focus on powering various parts of the internet with solar power. Perhaps the most notable is making the server for LowTech Magazine be completely powered by solar power [1]. This has sparked others to explore the constraints of solar power [29, 22]. One observed disadvantage of powering a server with solar power is that it becomes intermittent. *The Solar Protocol* is an attempt to mitigate this by placing solar powered servers across the globe that coordinate to serve the content from the server with the most power available [4]. Because it doesn't offer a way of opting out of the energy intensive infrastructure of the internet, the authors stress that they see the *Solar Protocol* as a "provocation to scrutinize, examine and explore ways of reducing [the growing energy impacts of the internet]" [4, p. 2].

2.2. Intermittency and users

Intermittency provide different interesting problems on the supply side, but what about the customers who use the services? The collection of studies in the previous section has a blind spot in how the intermittency of these services affect the people that use them, highlighting the need for further investigation.

One such area of inquiry is how being disconnected from the internet affects people's daily lives. In Widdicks et al.'s study, the participants were instructed to turn off connections to the internet on their devices, and only turn them on when they thought they needed to [31]. The participants reported an improved sense of focus and productivity, which the authors propose as an additional motivation (in addition to sustainability) for restricting internet use. It is worth noting that the design of the study centres around voluntary disconnection, where the participants have full control over when they are disconnected from the internet, whereas in a setting with an intermittent internet based on renewable energy, this would be outside of people's control.

There are also studies of intermittency that don't focus on the internet. Jensen et al.'s study on shifting laundry through scripting [16] studies how people experience intermittency in connection to laundry. A device called "the Box" is installed in the participants' homes. It alters the script of the washing machines — the "implicit user's manuals" that tell the user what normal interaction with it is — such that washing when it is windy becomes the default, the normal thing to do. Notably, it cuts off the power to the washing machine until there is enough wind or an emergency button is pressed. This makes it behave as if it is intermittent.

Laundry is an activity with a fairly narrow goal (to have clean clothes) whereas the internet can be used for many different purposes. Both wind and solar power's output vary with time, but output of the sun is more periodically predictable. There are in other words differences between the activity of laundry and internet usage — as well as between wind and solar power — that point to potential further investigation.

3. Method

This study is based on Critical Design. Critical Design is something that is less of a method with steps you can follow than an approach to research. As the name suggests, it takes from critical theory and design, aiming to question and understand assumptions better through design. This is typically done through a provocative design that through provocation makes an underlying assumption visible, and thus prompts us to think about it. It is a form of participatory research. The research is conducted through dialogue, where the researcher creates a design that hopefully provokes thoughts among the participants [25, 2]. The focus is less on gathering data from the participants, and more about letting the participants build their understanding and build theories of understanding around the subject.

3.1. Participants

At first I tried recruiting participants by posting a flyer text in various channels that were either sustainability focused or were connected to my study programme. That yielded zero responses. I was more successful with a convenience sampling approach, where I asked people that I already knew from my studies. That yielded four participants, all of whom were residing in Sweden, but P1, P3, and P4 were exchange students from other countries. Three of the people who declined participating were briefly interviewed in order to understand why they didn't want to participate.

	Household	Country	Demographics	Common internet habits
P1	H1	CH	M20-30, student in the IT department	studying, watching tv series, video calls, social meida
P2	H2	SE	M20-30, student in the IT department	programming, video, studying
P3	H2	CH	F20-30, student in the IT department	social media, video, studying
P4	H3	DE	M20-30, student in the IT department	music, video, video calls, studying, software development

Table 1: The demographics and internet habits. The country codes refer to country of origin.

	Country	Demographics
D1	SE	M20-30, student in the IT department
D2	CH	M20-30, student in the IT department
D3	SE	M20-30, software developer

Table 2: The demographics of the people who declined to participate when asked.

3.2. Design

The provocation is done using a travel router of the model GL iNet Gl-AXT1800 Slate AX. It runs OpenWRT and is programmable. It was installed in the participants' home and controlled the available bandwidth following an approximation of the strength of the sunlight. After the installation process the participants rarely interacted with it, as its primary function is to control the network traffic that is passing through. The participants were instructed in how to install and uninstall it, and were told that they are allowed to temporarily uninstall it when they feel the need to. For the purposes of this report I call it the Solar Router.

3.2.1. Bandwidth throttling

The emulation of the solar internet done in this project consists of throttling the bandwidth according to an approximation of the strength of the sunlight. This is done in two steps. First, it is assumed that the sun shines the brightest at noon and that it does not shine at all during the time between sunset and sunrise (i.e. at night). The strength of the sun between sunrise and noon is calculated with a linear interpolation. The data for when sunrise, noon, and sunset occurs is taken from `https://sunrise-sunset.org/api` . See fig. 1 for a visual representation.

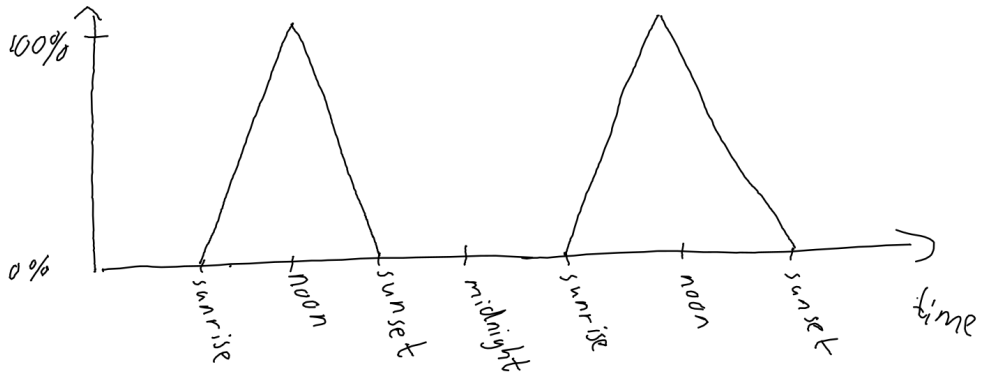


Figure 1: An illustration of how the sun's strength is approximated based on the time of day.

Secondly, the effect of the cloud on the power generated is also approximated using a linear interpolation. The efficiency of solar power is assumed to dip to 25% when overcast. The SMHI forecast API gives a forecast about the mean cloud coverage for a location.[1] When the sky is forecast to be maximally cloudy, the strength of the sun is approximated at 25%, rising to 100% when there is no cloud cover. The values in between are calculated with a linear interpolation. See table 3 for the precise values.

tcc	0	1	2	3	4	5	6	7	8
sun	100%	91%	81%	72%	63%	53%	44%	34%	25%

Table 3: How much the bandwidth is affected by the total cloud coverage.

These approximations are then combined into a single percentage value of the maximum bandwidth. The bandwidth cap is updated every 15 minutes. An example of how the bandwidth ceiling might look like can be found in fig. 2.

This study was conducted in Sweden during the autumn and spring, when the sun sets fairly early at around 5 p.m. Because of this, participants who have a regular day job will

[1] `http://opendata.smhi.se/apidocs/metfcst/parameters.html`

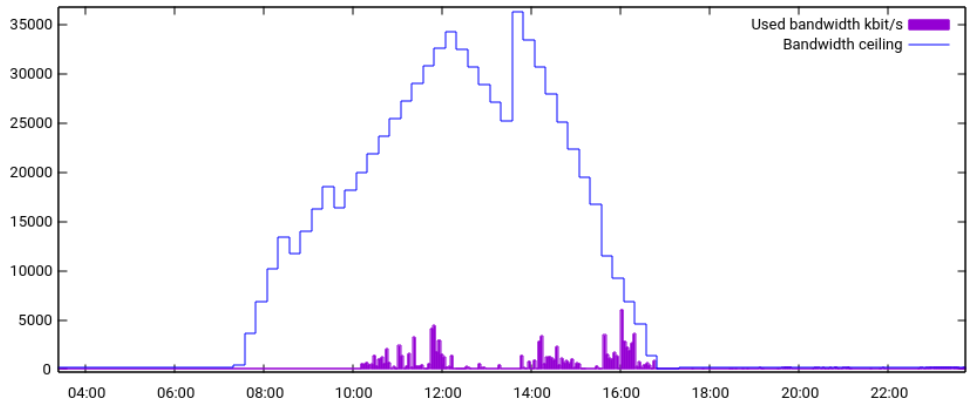


Figure 2: An example trace of the dynamically set bandwidth ceiling and the amount of bandwidth used.

most likely spend their time at home after the sun has set. With a strict interpretation of an internet powered by the sun, this would mean that the participants have no internet at all at home. To make it a bit more dynamic, and to not completely disconnect the participants, I decided to add a battery backup to the setting. When the sun has set, the battery backup is engaged which gives a bandwidth of 1Mbit/s per person in the household and lasts for 300 MByte/person. This yields 40 minutes of use time per person at maximum continuous bandwidth usage. When the battery is depleted the bandwidth cap is set to 100kbit/s in order to never completely cut off the participants from the world. Both the bandwidth and the quota are shared between all members of the household. These numbers were chosen in order to be provocative, and for that "[a] slight strangeness is the key — too weird and [the changes] are instantly dismissed, not strange enough and they're absorbed into everyday reality" [8, as cited in, 2, p.294]. I aimed for parameters that would make the limitations noticeable but not harsh enough that the participants would dismiss it.

3.2.2. Technology used

- bandwidthd-sqlite: `https://openwrt.org/packages/pkgdata/bandwidthd` `https://openwrt.org/docs/guide-user/services/network_monitoring/bandwidthd`
- cron
- my own script
- tc-netem
- the `bandwidth_used.sh` script from `https://github.com/RVgo4it/OpenWrt/tree/c789dea879133f60d9516cdaf6456512f9281ff1/`

- smhi api
- sunrise api
- gnuplot to make the plot from the database

In this section I describe the technology used to make the intervention. I used a program called `bandwidthd` (specifically the `bandwidthd-sqlite` package in Openwrt) to get quantitative data on the participants' usage patterns, which served as a jumping-off point for the interviews. It monitors the bandwidth usage per device over time. It can produce its own graphs, but I use its database to produce graphs that also contain data from my own script. The database units of the database aren't documented, so the graphs I have produced have been adjusted manually by comparing the graph produced by `bandwidthd`. This makes them imprecise, but for the purposes of serving as a base for the interviews, this level of accuracy should be sufficient.

The strength of the sun is approximated using SMHI and Sunrise's APIs (see section 3.2.1). This approximation is then used to create the network conditions that are this intervention. It is divided into two parts. When the sun is shining the bandwidth is throttled to be proportional to the strength of the sun using the Linux traffic control subsystem (through the command `tc`), more specifically the `netem` module that allows for emulating various network conditions. `tc-netem` is used to create limits on the bandwidth (both incoming and outgoing traffic).

The quota is implemented using the `bandwidth_used.sh` script that tallies the entries in the database and checks if a quota has been reached. If the quota has been reached for a given night, the bandwidth is further limited down to 100 Kbit/s/person for the rest of the night using `tc-netem`.

This is all controlled by a script written by me (see section A.1) that is run every 15 minutes using the `cron` utility. It downloads the latest data from the APIs and updates the bandwidth according to what is written above. The relevant numbers (time of day, available bandwidth, whether the quota has been exceeded) are logged to a file so that they can be graphed (using gnuplot) together with the bandwidth data from `bandwidthd`.

3.3. Limitations

The strength of the sun is only approximated, and is thus not very accurate. It is updated every hour, and is based on forecasts and crude interpolations instead of actual measurements. For the purposes of this study, this doesn't matter, because it would not be possible to accurately emulate every change that would happen if the internet was re-engineered to only run on solar power (see [24] for an attempt at mapping the challenges an intermittent-energy internet architecture would entail). There would simply be too many systems involved, most of which I don't have any control over. For example, servers would also run on solar power, and if the server is in a region without sun, that

server would be inaccessible. So instead of trying to be accurate, I focus on creating an environment where connection is intermittent and bound to the motions of the sun.

SMHI, the Swedish Meterological and Hydrological Institute, has an API for measuring the amount of sunlight an area receives per hour[2] , but the API was too slow for the purposes of this project, sometimes taking several hours to publish the data. I also considered using live data from solar panels owned by my supervisor, but the manufacturer's API seems to only publish net power production data (i.e. power generated by the solar panels subtracted by the power consumed by the household), and not the gross power production data necessary for this project. Further, the real time API allows only two connections, making it more complicated to have multiple instances of the study running concurrently. The alternative I settled for was SMHI's forecast of cloud coverage and an API for sunsets/sunrises along with some linear interpolation. The reason being that it was the easiest alternative while being reasonably real-time, at the cost of accuracy.

3.4. Data collection: trace interviewing

The intervention was conducted over two weeks for each participant. The first week was an observation phase where nothing about the participant's network condition was changed. The Solar Router was just installed to do some basic observation in order to establish a baseline to compare to when the intervention part started. The data recorded was bandwidth usage for the entire household over time. The second half of the study was the actual intervention part. This was when the bandwidth throttling described above took place.

After each week a semi-structured interview was conducted with the participant. The first interview tried to get a picture of the participant's normal internet usage, what devices they have, what services they use, how the services are used throughout the day, and what they are used for. The second interview took the intervention as a jumping-off point and sought to understand how their daily practices changed, what problems they encountered and if they managed to work around them, and if they perceived any benefits from the intervention. The interviews were transcribed and analyzed manually, and were main source of data for this study. The bandwidth monitoring served as a jumping off point for the interviews, the qualitative interview data was the focus of my analysis. This approach was inspired by [11].

[2]The parameter "Solskenstid" at: `http://opendata.smhi.se/apidocs/metobs/parameter.html`. There is an example demo at `http://opendata.smhi.se/apidocs/metobs/demo.html`.

4. Results

4.1. Notable changes in behaviour

All participants said they worked less from home, and in general were less active when the sun had gone down. P1 was in general the most positive towards this. He would go home later and stay at campus longer to do work since there was an internet connection there. This was both good and bad, since he got more work done but he also didn't want to overwork himself. He expressed that spending less time on social media and videos was a healthy change for him. At home, he replaced much of his internet activity with reading books or spending time with friends. He also mentioned that he wanted to focus more on cooking and spend more time on meditation, which he thought the solar internet could help with if the study had continued.

P4, P2, and P3 were more negative. They had a habit of doing work (here defined as anything that they felt productive, including studying but also personal projects) from home, which was harder to do with the solar internet. Often an internet connection was necessary to do research, or communicate with others involved with the same project. The ability to do such activities were limited, and especially when the battery had run out such activities were deemed infeasible. P4 replaced this with some reading and sewing, which he felt positive towards to a degree, but also said that "it's also one of these things which you cannot do for hours right, so um I would have liked to,- I'm partially happy that I kind of had some time for it but this was too much time" and "there is a difference between voluntary chilling and you kinda don't have any other choice". Even though P4 thought these changes were negative on the whole, one change he liked was an improved sleep schedule. Even one week after the study, he would get into bed one or two hours earlier every day.

P2 and P3 replaced a lot of their internet activity with mobile games. Since much of that internet activity was time they felt were productive, they felt this was a net negative. P2 noted that there are two kinds of internet activities, those that are plannable and those that are not. Plannable activities are those that can be rescheduled, or done in advance. This is an example of a participant-made theory described in section 3. A good example is downloading a TV series to watch later. With other activities you can't do that, because the activity requires a connection to perform. Examples of this include phone calls, but also research as you often don't know what you want to look for beforehand. Research wasn't done purely for the sake of work either. P2 and P3 would often look things up during conversations, and they felt that practice was disturbed with the slow internet. For P4, P2, and P3, work related activities tended to fall into the unplannable category and leisure activities tended to be plannable. One notable exception is phone calls over the internet. P4 had to hang up a call earlier than he would have liked because he was afraid the connection would become too bad for having a conversation. P3 said that before, she didn't need to use the internet for studying, and wasn't even allowed to use it during weekdays at her boarding school. But now she feels like she can't study without the internet. All three felt they were doing less work than before, while spending more

time being bored. For them, this was an undesirable outcome because they felt it cut into valuable work time. In the long run they felt this would not have been sustainable.

P2 remarked that one problem with the setup of only having internet for part of the day is that the internet becomes a limited resource that is used for both work and leisure. For example, P2 worked less during the mornings because he was happy to get his internet connection back, making it possible to watch videos again.

4.2. Perceived problems with software

One throughline among the participants is that they thought the software and services they regularly use aren't sufficiently adapted to the intermittent environment. Most notably, streaming services rarely provide an adequate option to download the content. P2 and P3 had a habit of watching a TV series on Netflix during dinner. Since that usually happened after the sun went down the ability to stream was limited, but they could not find an option to download the show on the TV. Instead they downloaded the TV series on a phone (where the option was available). Because of some anti-piracy feature they weren't able to use Google Cast to display the show on the TV. P2 thought it was satisfactory to watch on the phone, but P3 disagreed and said the subtitles were too small. P1 said he tried to download a show but the show in question wasn't available for download. This isn't always a technical issue, but a legal one. For example, Netflix states that some titles can't be downloaded because of licensing agreements.[3]

P4 did manage to download a lot of content, but expressed frustrations with how the software worked, in particular with Spotify. He uses it to download podcasts, but said that he downloaded so much that "it's getting annoying to press the download button and then remove it from the downloads again on Spotify". The podcast episodes wouldn't get removed after being played, which took up storage space and required manual deletion. He also noticed that sometimes Spotify would refresh its cache, even though he hadn't changed which episodes should be downloaded. It would download some things again, without consideration to time of day or how much bandwidth was available. When he noticed that he rushed to close the program to save on precious bandwidth. That was "a miserable experience!".

P4 tried to use Overleaf (a cloud based word processor) one day, because he needed to work on a upcoming deadline. He said it was unusable unless all other browser tabs were closed.[4] The experience was ok, but he had to be extremely intentional about it. What he did was that he decided to close down everything that could consume bandwidth, and open only Overleaf for the entire evening.

Some websites worked better than others when the battery had run out and the available bandwidth was 100 kbit/s per person in the household. P2 browsed a lot on `reddit.com`

[3] `https://web.archive.org/web/20231223103931/https://help.netflix.com/en/node/54870`
[4] This is not expected from the study design. See section 6.1 for details.

and `news.ycombinator.com` because those are very light-weight and feature mainly text. P3 on the other hand liked to browse social media, which didn't work well due to their greater bandwidth usage.

4.3. Reasons for not participating

I interviewed three people who declined to participate in the study, seeking to understand why. For these interviewees, obligations or other relationships to other people were the main reasons for declining.

D1 said that the main reason he declined was that he felt he constantly had to be online. This was jokingly exaggerated, but he then explained that his job requires him to be constantly available online. Various customers or business partners could contact him on various platforms at basically any point in time, and so he would need an internet connection even in the evenings. The added friction would have been too much. If it wasn't for that job he would have considered participating.

D2 was playing online multiplayer games with friends every evening. She felt like she would have had to unplug the Solar router every day, and that that would make for bad data. I had insisted that it would still be very interesting for the study, to which she responded that it felt like it would be have been less of a waste of time to have some other participant that would be able to follow the rules of the Solar router. Abstaining from games was not an option as they are her best method of winding down and an opportunity to socialise with friends. Disconnecting during a game can be disruptive, but in Dota 2 the teammates can pause the game so that you can attempt to reconnect, so it is still manageable. She was more worried about voice calls as they can be unpleasant with a poor connection.

D3's reason was his custom network setup. Adding a device that would make the internet very slow would make the setup more complicated and the process of diagnosing problems more annoying. He anticipated that participating would entail much extra work that he wasn't keen on doing. On top of that, he maintains a couple of home servers, some of which he hosts for relatives. This brings extra responsibility, troubles with the network wouldn't just affect himself, but also those he provides a service to. Sometimes the home server becomes unreachable, which he usually learns about from a text message from a relative. To fix the problem usually takes the better part of an afternoon.

Both D1 and D2 felt that they would need to be able to abstain from the internet in order to contribute meaningfully to the study. This was despite me insisting that it would be interesting for the study even if they disable the router every day.

4.4. Summary of changes

Participant	Change
P1	Stayed at campus more to use internet, e.g. for videos.
	Read more books.
	Stopped using social media during evenings.
P2 & P3	Temporarily uninstalled in order to watch a TV program with an invited guest.
	Stopped watching (internet-connected) TV during meals.
	Switched to watching a downloaded show on P4's phone.
	Stopped watching films.
P2	Read more lightweight websites like Hackernews and Reddit.
	stopped working in the mornings
P3	Played more mobile games.
P4	Downloaded music and podcasts.
	Played more mobile games.
	Stopped having voice calls for studying.
	Watched less Youtube, including educational videos.
	Used translation services less.
	Rescheduled one voice call from the evening to the day.
	Temporarily uninstalled in order to have a voice call.
	Stopped doing open source development.

Table 4: Summary of notable changes the participants made during the intervention.

5. Discussion

5.1. Intermittent laundry and internet

There is an interesting contrast between the responses from my participants and the participants in Jensen et al.'s study [16]. To recap, their study is about shifting laundry to when there is sufficient wind power. The participants had a prototype installed in their home called the Box that would turn off power to the washing machine at times of insufficient wind.

From what I can tell from the paper, most of the participants who interacted with the Box were more positive towards their intervention. They saw it more as a game or as a tool, while my participants expressed more negativity towards the Solar Router.

One notable difference between the method of the two studies is the level of interactivity and feedback the two designs provide. The red-and-green clock of the Box turned the Box into a game Sam, one of the participants, wanted to stay ahead of. The Box also has a display that shows a score of sorts, how many times the family has pressed the override

button that lets them wash regardless of the wind. The Solar Router on the other hand didn't have anything tangible to interact with. It was possible to plan ahead, but the lack of any feedback made it hard to know when the bandwidth would become too low. This could happen in the middle of an activity, say a video or a phone call. Similar complaints about lack of transparency have been made by people living with monthly bandwidth caps in South Africa [5]. P4 for example said that he hung up a video call earlier than he would have liked because he didn't know when the call would cut off due to the poor connection. He would rather hang up early and end the conversation properly instead of being cut off. With the Solar Router, the internet was available until it was denied. The Box on the other hand had a display with a forecast of when there would be enough wind so the participants could plan ahead. It could be difficult to find a "green" time slot, but once you found one you were not denied it.

Both designs had an escape hatch that participants could use to temporarily bypass the limitations imposed. With the Solar Router, the participants were instructed that they were allowed to unplug it when they felt it was necessary. The Box had a button that temporarily allowed the participants to wash, regardless of how windy it was. While it is technically possible to rewire some ethernet cables, the design of the Solar Router doesn't convey that it is something you are allowed to do during normal use. The Box on the other hand, had a button for the bypass system, making it clear that it is part of the intervention. It was clearer how you were supposed to interact with the Box, it had a clearer script.

Another important difference is the different characteristics of the activities involved. Laundry is an activity that is characterised by the need of clean clothes. That need can be temporarily fulfilled. The internet on the other hand is used for many purposes: work, leisure, or communication. Washing turned out to be shiftable to a significant degree, both through washing at different times and through washing less. Much internet activity was also shiftable, for example through downloading content in advance or replacing the activities with offline ones, but some participants (P2, P3, P4) also felt that it interfered with their work routines.

Torriti et al.[30] state that high energy forms of leisure such as watching TV tend to be less flexible (citing [12]). Domestic cleaning practices on the other hand tend to be more flexible (citing [15]). This rhymes well with both my and Jensen et al.'s findings. The participants seem to be more positive towards being more flexible with laundry than with the internet. It is not a perfect correspondence however, as the leisure activities within internet use were often more flexible than work activities. Or, rather, leisure activities were more easily pre-downloaded or replaced with other offline activities.

A third important difference is that solar power is predictable in a negative way: the time people got home was also the time the sun was setting. The wind on the other hand is more unpredictable, but it can be windy at any time of day, meaning that laundry could be done at preferred times of day if they were lucky enough.

5.2. Peak loads

Network operators plan their infrastructure based on peak demand [27], meaning that limiting peaks is important to limit the expansion of infrastructure. In 2019, video made up for 60% of traffic on the internet [28]. At the same time, video is a type of traffic that concentrate in peaks, which means that it has a higher ratio between peaks and average traffic compared to file sharing [6]. This means that video is a prime target for shifting. Turning streaming into downloading and spreading it out throughout the day could bring down peaks in internet traffic.

My study limited access to the internet during the peak hours, and indeed, all of my participants reported that they streamed less video during the evenings. All of them tried to shift their bandwidth usage by downloading during the day, but were hindered by licensing, technical restrictions enforced through law (DRM), or interfaces that weren't well suited towards downloading. Even though they didn't always succeed, they at least showed a willingness to try. This is corroborated by Widdicks et al.'s study, where it was concluded that entertainment often was among the least important activities performed through the internet, meaning they were the easiest to remove or replace with an offline alternative [31].

5.3. Who can shift

The internet is a relatively new invention that didn't become ubiquitous until even more recently. It's easy to assume that we can revert back to the days without the internet, but Kratel challenges that assumption. In asking who can or cannot disconnect from the internet, she argues that the ability to disconnect is tied to various privileges, such as gender and class [18]. Since obstacles to disconnection may not be obstacles to shifting, it is unclear how the dynamics of privilege translate over, and as such I believe it is necessary to investigate this question further. In this section I discuss potential factors that can affect people's ability to shift.

Figueiras et al. argue that that disconnection is often framed as a question of individual agency, but is in reality a collaborative effort. This was apparent in the answers from the interviews in this study (see section 4). I tried to make it easy to participate for the individual participants, but neglected the social relations they might have that hinders disconnection. First, work is a social obligation that can be hard to negotiate, creating effectively hard boundaries to shifting. D1 declined because his obligation to business partners and customers made it hard to voluntarily disconnect. It was specifically the need to maintain connections to others that informed this choice.

The people who participated in the study were all students who were free to choose when to work, but even then the Solar Router still interfered with their ability to do so. The ability to disconnect isn't just a collaboration with other people, but also with the software systems that you depend on. P4 for example had his thesis on Overleaf, which

meant that he could only write on it when he had a sufficiently good connection. Similarly, many participants in Widdicks et al.'s study said using Google Docs for collaborative writing was necessary for their work [31]. Cloud based real-time collaboration software have their benefits through the collaboration it enables, but it also creates a dependency on a connection. This is not always a personal choice, as a collaboration requires that all involved parties agree on a particular approach. Local-first software [17] could be one way to make this type of real-time collaboration software work through intermittent connections.

In section 4.1 I observed that leisure activities tended to be plannable for the participants. However, no participant played online games during this period. This is a significant blind spot since it can be difficult to play online multiplayer games with a poor connection. Many online games are by their nature unplannable, since the internet connection is used to communicate and interact with other players in real-time. D2, however, was more interested in the social aspect of playing games. This highlights the role in which the internet connects people.

Sometimes, disconnection is framed in terms of personal experience, to not directly use an internet connected device. But as the case of D3 shows, disconnection in the home can have different forms of consequences, and might even force an active intervention to investigate what is or isn't wrong with the home sever. The individual might be doing an offline activity when the household's connection goes down, which produces the need to go and check that the server and the network is still functioning correctly.

D3's discomfort points the way to an open design space. While there have been several attempts at creating a website running on solar power [22, 29], the possibility of home-servers or remote file storage remain an unexplored area in the Solar Internet. There are several open questions. How does one deal with the intermittency? Is it possible to make something like the Solar Protocol [4], where several servers across different time zones cooperate?

D3's response also shows that the distinction I have made in this study between servers and end users is a simplification. The internet is not just a set of clients, there are also servers and network infrastructure. Others have studied the impacts of intermittency on servers [22, 1, 29] and others have studied its impacts on networks [24, 20], but since those are not disconnected components it will eventually be necessary to study them together in an integrated system in order to have a full picture.

5.4. Parameter tuning

Jensen et al. write that designers "sometimes also unintentionally design for imaginary actors with specific values, norms and ethics" when they "attempt to script specific behaviours into designs" [16, p. 1395]. This happened to me as well. My imagined actor was heavily based on myself, it was an actor who does not perceive a need — personal or

social — to use a lot of bandwidth during the evenings. The services it regularly was using were already bandwidth light, and those that were not had the option for downloading the content in advance. Crucially, this actor didn't work in the evenings, and had satisfactory means of leisure that worked offline. This influenced the values of the parameters for the battery and bandwidth caps.

This did not match well with the participants, their expectations were different. P2, P3, and P4 expressed surprise and frustrations at how slow the internet became at night. In the literature, there is some consensus that too much strangeness or provocation can cause participants to reject the design [3, 2]. The people who declined to participate (see section 4.3) could be seen as an instance of this. But it is also interesting that while the change was very harsh to the people who did participate in this study, it was also something that was hard to dismiss, as it was more or less imposed on them. Internet use was part of their daily habits, so it wasn't easy to choose to not interact with the provocation — non-interaction with the design was also a big change in behaviour that was caused(?) by the provocation. They were encouraged to temporarily uninstall the Solar Router if they needed to, but again, doing so would also be interacting with the design. The only way to outright dismiss the design was to cancel the study all-together.

With that said, even if the participants did not outright dismiss the design, I didn't succeed in creating the strangeness I set out to create. The aim was to put in more friction when using the internet, gradually degrading the quality until the internet was effectively cut off. I wanted there to be more fidelity in the intervention than that the internet turned off at sunset. One salient pattern from the collected bandwidth usage data was that the bandwidth usage seldom reached the bandwidth ceiling during the day (see e.g. fig. 2). It was only close to sunrise and sunset that they were likely to hit the ceiling in a way that was noticeable. The parameters that are relevant for tuning the intervention are:

- the amount of bandwidth that is available *with* the battery
- the amount of bandwidth that is available *without* the battery
- the size of the battery data quota

As Stojanov et al. note, the constraints posed by the Solar Internet are elastic and flexible, inviting negotiation between different sets of values [29]. How these parameters should be tuned is not given, as it becomes a trade-off between comfort and the resources required to manufacture the batteries.

6. Future work

6.1. Inconsistent throttling

One problem with the implementation was that it wasn't consistent. One participant (P4) experienced a stronger throttling than expected. For example, when the sun had gone down, cloud based text editors like Overleaf were not usable if he had other browser tabs (with calendars or chat applications) open. After the intervention had ended P4 tested the router again by installing it in his home. This time it worked as expected. He had moved to a new home during the time between the intervention and the second test, which suggests the problem might manifest only in certain network conditions. I advise future researchers who want to do a similar study to develop a routine for checking for problems when installing the router.

Another issue occurred with P1. In the middle of the night he tried to transfer photos between one device to another. This should have been very slow because it happened during the night, but the recorded bandwidth usage during that time far exceeded the bandwidth limit. It might be that `tc-netem` can't handle very high bandwidth usages.

Some routers have an option to enable hardware acceleration. This interferes with `tc-netem`, causing it to not function at all.

6.2. Better predictability

As written above, the participants said the Solar Router was unpredictable. It was not clear when the battery would run out and the connection between the sun and how much bandwidth was at times opaque. To improve on this one could update the bandwidth settings more frequently, and add an interface with a forecast like in [16].

There is also room for have a clearer intention behind the parameters. See section 5.4.

6.3. The role of quantitative data

The quantitative data has been useful, but mostly for monitoring the router itself, to see if it is behaving as expected. It hasn't provided much insight into the participants' experience, and provided only a few minor jumping-off points for questions. The main problem was that the data was too coarse-grained. The idea was to point at interesting patterns in the data but since it doesn't differentiate between devices or services used it was hard for the participant to remember what activity corresponds to a certain network activity. This points to the need for more sophisticated techniques for classifying network data that can identify both end user devices and what service is in use. The participants found it hard to remember what they were doing at a specific point in time, so any additional information such as a specific service would be helpful.

7. Conclusion

In order to provoke assumptions around connectivity, I disrupted four participants' experience with the internet by making sure that they did not have full access to the internet all the time. Their internet access was severely limited when the sun was not shining, making the internet a service that is fully accessible only during parts of the day. This resulted in the documentation of various specific experiences of adapting, as well as some increased understanding of what can make it harder to adapt to intermittency. In addition to that, this has been an exploration of an approach to challenge the Cornucopian paradigm from the side of limiting demand, with lessons learned for how to continue this line of research.

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A. Appendices

A.1. Throttling script

```ruby
#!/usr/bin/env ruby
require 'uri'
require 'net/http'
require 'json'
require 'time'
require 'English'


class APIs
  attr_reader :smhi_data, :sunrise_data
  def initialize(long, lat)

    @long = long
    @lat = lat


    @smhi_data = JSON.parse(
      Net::HTTP.get('opendata-download-metfcst.smhi.se',
                    "/api/category/pmp3g/version/2/geotype/point/"\
                    "lon/#{@long}/"\
                    "lat/#{@lat}/data.json"))

    data = JSON.parse(
      Net::HTTP.get('api.sunrise-sunset.org',
                    "/json?lat=#{@lat}&lng=#{@long}"\
                    "&date=today&tzid=Europe/Stockholm"))

    @sunrise_data = {
      sunrise: Time.parse(data['results']['sunrise']),
      noon: Time.parse(data['results']['solar_noon']),
```

```
      sunset: Time.parse(data['results']['sunset']),
    }

  end

end

def get_tcc(json)
  parameters = json['timeSeries'][0]['parameters']
  for p in parameters
    if p['name'] == 'tcc_mean'
      tcc_mean = p['values'][0]
    end
  end
  tcc_mean
end

def interpolate(ystart, ystop, xdist, x)
  k = (ystop - ystart)/xdist
  k * x + ystart
end

def get_sun_strength(apis, now)
  sunrise = apis.sunrise_data[:sunrise]
  noon = apis.sunrise_data[:noon]
  sunset = apis.sunrise_data[:sunset]

  if sunset <= now
    0 # interpolate(0, 0, next_sunrise - sunset, now - sunset)
  elsif now <= sunrise
    0
  elsif sunrise <= now && now <= noon
    interpolate(0, 1, noon - sunrise, now - sunrise)
  elsif noon <= now && now <= sunset
    interpolate(1, 0, sunset - noon, now - noon)
  else
    throw 'Something went wrong!  Should not happen!'
  end
end


def calculate_sun_cover(apis)
  interpolate(1.0, 0.25, 8, get_tcc(apis.smhi_data))
end
```

```
def calculate_bandwidth(apis, max_kbit, now, min_kbit: 0)
  [min_kbit,
   (get_sun_strength(apis, now) * calculate_sun_cover(apis) * max_kbit)
     .floor].max
end

class Throttler
  def initialize(device1, device2, device3, apis, log_file: './bandwidth.log',
                 max_bandwidth_kbit: 100_000,
                 min_bandwidth: 1000,
                 bandwidth_quota: 300, # Mbyte
                 exceeded_quote_bandwidth:  300 # kbit/s bandwidth to use when
                                                # the quota has been exceeded
                )
    @max_bandwidth_kbit = max_bandwidth_kbit
    @min_bandwidth = min_bandwidth
    @bandwidth_quota = bandwidth_quota # Mbyte
    @exceeded_quote_bandwidth = exceeded_quote_bandwidth # kbit/s bandwidth to
                                                         # use when the quota
                                                         # has been exceeded

    @device1 = device1
    @device2 = device2
    @device3 = device3
    @apis = apis
    @bandwidth_log_file = log_file

    @throttled_bandwidth =
      begin
        IO.read('last_bandwidth').to_i
      rescue Errno::ENOENT => _e
        false
      end

    @exceeded_quota =
      begin
        IO.read('./last_bandwidth_quota_hit').to_i
      rescue Errno::ENOENT => _e
        false
      end

  end

  def quota_to_graph_value
```

```
  if @exceeded_quota then 100_000 else 0 end
end

def throttle(now = Time.new, dummy: false)
  `tc qdisc add dev #{@device1} root netem rate 1gbit` unless dummy
  `tc qdisc add dev #{@device2} root netem rate 1gbit` unless dummy
  `tc qdisc add dev #{@device3} root netem rate 1gbit` unless dummy

  IO.write(@bandwidth_log_file,
           "#{now.to_i},"\
           " #{foo(@throttled_bandwidth || 0, dummy)},"\
           " #{quota_to_graph_value}\n",
           mode: 'a')

  @throttled_bandwidth = calculate_bandwidth(@apis, @max_bandwidth_kbit, now)

  if get_sun_strength(@apis, now).zero?
    sunset = @apis.sunrise_data[:sunset]
    puts "sh bandwidth_used.sh -d --quota=#{@bandwidth_quota} "\
         "-h=#{sunset.hour} --bytes=m -s=1"
    puts %x(sh bandwidth_used.sh -d --quota=#{@bandwidth_quota}
          -h=#{sunset.hour} --bytes=m -s=1)
    if $CHILD_STATUS.exitstatus == 250
      IO.write('./bandwidth_quota.log',
               "#{now}: Exceeded quota of #{@bandwidth_quota}."\
               " Now throttling to backup bandwidth of "\
               "#{@exceeded_quote_bandwidth}\n",
               mode: 'a')
      IO.write('./last_bandwidth_quota_hit', now.to_i.to_s,
               mode: 'w')
      @throttled_bandwidth = @exceeded_quote_bandwidth
      @exceeded_quota = true
    end
  elsif File.exist?('./last_bandwidth_quota_hit')
    File.delete('./last_bandwidth_quota_hit')
  end


  unless dummy
    `tc qdisc change dev #{@device1} root netem rate #{@throttled_bandwidth}kbit`
    `tc qdisc change dev #{@device2} root netem rate #{@throttled_bandwidth}kbit`
    `tc qdisc change dev #{@device3} root netem rate #{@throttled_bandwidth}kbit`
  end
  IO.write(@bandwidth_log_file,
```

```
              "#{now.to_i}, #{foo(@throttled_bandwidth || 0, dummy)},"\
              " #{quota_to_graph_value}\n",
              mode: 'a')

    IO.write('last_bandwidth', @throttled_bandwidth.to_s, mode: 'w')
  end
end

def foo(bandwidth, is_dummy)
  if is_dummy
    "0, #{bandwidth}"
  else
    "#{bandwidth}, 0"
  end
end

if __FILE__ == $PROGRAM_NAME
  now = Time.new
  now = Time.parse(ARGV[0]) if ARGV.length == 1

  persons = 2 # The number of people in the household
  max_bandwidth = 100_000
  min_bandwidth = persons * 1000
  bandwidth_quota = persons * 300
  exceeded_quote_bandwidth = persons * 100

  device1 = 'eth0'
  device2 = 'eth1'
  device3 = 'eth2'

  begin
    if ARGV.length == 2
      Throttler.new(device1, device2, device3, APIs.new(17.64805, 59.85804),
                    max_bandwidth_kbit: max_bandwidth,
                    min_bandwidth: min_bandwidth,
                    bandwidth_quota: bandwidth_quota,
                    exceeded_quote_bandwidth: exceeded_quote_bandwidth
                   ).throttle(now, dummy: Time.now < Time.parse(ARGV[1]))
    else
      Throttler.new(device1, device2, device3, APIs.new(17.64805, 59.85804),
                    max_bandwidth_kbit: max_bandwidth,
                    min_bandwidth: min_bandwidth,
                    bandwidth_quota: bandwidth_quota,
                    exceeded_quote_bandwidth: exceeded_quote_bandwidth
```

```
                    ).throttle(now)
    end
  rescue => e
    IO.write('./smhi.log', "#{now}: crashed with error:\n#{e.backtrace}\n", mode: 'a')
    raise
  end
end
```